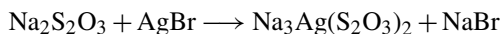
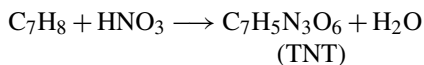


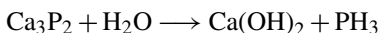
12. Aluminum sulfate has been used as a leather-tanning agent and as an antiperspirant. The compound can be obtained, along with water, by the reaction of aluminum hydroxide with sulfuric acid.
- Write a balanced equation for the reaction.
 - Assume that 100 g of aluminum hydroxide reacted with 155 g of sulfuric acid. What is the limiting reagent?
 - How many grams of aluminum sulfate are produced?
 - How many grams of the excess reactant are left at the end of the reaction?
13. One of the chemical reactions involved in photographic film development is the reaction of the excess AgBr with sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.



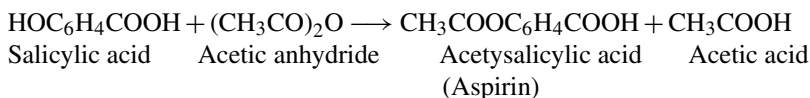
- Balance the equation for the reaction.
 - If 15 g of $\text{Na}_2\text{S}_2\text{O}_3$ and 18 g of AgBr are present in a reaction, which is the limiting reagent for the formation of $\text{Na}_3\text{Ag}(\text{S}_2\text{O}_3)_2$?
 - How many moles of the excess reagent are left at the end of the reaction?
14. The explosive trinitrotoluene (TNT) can be prepared by reacting toluene with nitric acid:



- Balance the equation.
 - Determine the limiting reagent if 65 g C_7H_8 reacts with 50 g of HNO_3 .
 - What is the theoretical yield of TNT that can be obtained from a reaction mixture that contains 65 g of C_7H_8 and 50 g of HNO_3 ?
 - If the actual yield of TNT for the reaction mixture in part (c) is 55 g, calculate the percent yield of TNT for the reaction.
15. Phosphine, PH_3 , can be prepared by the hydrolysis of calcium phosphide, Ca_3P_2 :

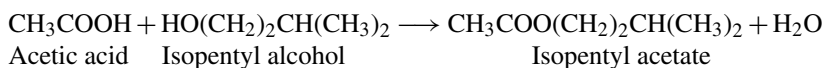


- Balance the equation.
 - What is the maximum mass of PH_3 that could be prepared by mixing 44.25 g of Ca_3P_2 and 125.0 mL of water? (Density of water at 25°C is 1.000 g/mL.)
16. Acetylsalicylic acid (the chemical name for aspirin) helps to relieve pain and lowers body temperature. Industrially, it is produced from the reaction of salicylic acid and acetic anhydride:

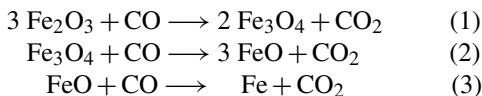


Starting with 12.00 g of salicylic acid and excess acetic anhydride, a chemistry student synthesized aspirin in the laboratory and reported a yield of 91%. What was the student's actual yield of aspirin?

17. The main chemical component responsible for artificial banana flavor is isopentyl acetate. An industrial chemist wants to prepare 500 g of isopentyl acetate by the reaction of acetic acid with isopentyl alcohol:



- How many grams of acetic acid are needed if the chemist expects a percent yield of only 78%? Assume that isopentyl alcohol is present in excess.
18. An analytical chemist wanted to determine the composition of copper-bearing steel. She placed 5.00 g of the sample in excess aqueous HCl solution. All the iron in the steel dissolved as iron (III) chloride, and 0.15 g of hydrogen was produced. What is the percent composition of the alloy? Assume the alloy is composed only of Cu and Fe.
19. A drug-enforcement agent seized 250.00 g of heroin, $\text{C}_{21}\text{H}_{23}\text{O}_5\text{N}$, and burned a 25.00-g sample in 35.25 g of oxygen. The end products were CO_2 , NO_2 , and H_2O .
- Write a balanced equation for the combustion reaction.
 - What is the limiting reagent?
 - How many grams of the excess reactant remain unreacted?
 - What was the percentage completion of the combustion?
 - Determine the total mass of CO_2 , NO_2 , and H_2O produced in the combustion process.
20. Iron is produced by carbon monoxide reduction of iron ore, usually hematite, Fe_2O_3 , in a blast furnace reactor. The following series of three reactions occur in various temperature regions in the reactor, resulting in the formation of molten iron.



How many kilograms of iron can be produced from 25.00 kg of the hematite ore? Assume that carbon monoxide is present in excess. (Hint: Manipulate these equations such that FeO and Fe_3O_4 cancel out of the final equation).